

Nicolò Bonacorsi

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Education

Columbia University MS in Applied Mathematics GPA: 3.6/4.00 <i>Teaching Assistant – APMA E4300 Numerical Methods</i>	New York, NY Aug 2025 – Dec 2026
Sapienza University of Rome BS in Mathematical Sciences for Artificial Intelligence , GPA: 3.99/4.00 <i>Final Grade: 110/110 cum laude</i>	Rome, Italy Sep 2022 – Sep 2025
University of Bologna BS in Statistical Sciences , GPA: 3.95/4.00 <i>Final Grade: 110/110 cum laude; Award: University of Bologna Prize for Outstanding Student</i>	Bologna, Italy Sep 2021 – Jul 2024

Research & Publications

Columbia University <i>Supervisor: Professor Ioannis Karatzas</i> <i>Graduate Researcher</i>	New York, NY Dec 2025 – Present
- Derived a tractable one-dimensional free-boundary formulation for a controlled diffusion with jointly optimized drift and volatility; characterized optimal state-dependent feedback policies and endogenous stopping thresholds. - Established regularity and verification conditions guaranteeing optimality for finite-cost feedback controls, including degenerate zero-volatility policies outside classical smooth-verification settings.	
Northeastern University – MENTIS Laboratory <i>Supervisor: Professor Francesco Restuccia</i> <i>Research Intern</i>	Boston, MA Dec 2025 – Present
<i>Credal Rate–Distortion Theory – IEEE ITW, 2026. N. Bonacorsi, M. Caprio, F. Meneghello, F. Restuccia</i> - Formulated a rate–distortion framework for robust ML inference under epistemic uncertainty and model misspecification, preserving distributional ambiguity through compression. - Proved that zero-distortion credal coding reduces to graph-based combinatorial optimization and built an LP/graph-coloring pipeline solving 54,000 set-membership programs; compressed 60 SVHN credal sets into 14 codewords (4.29× with zero incompatibility violations versus 18.6% for an equal-size barycentric baseline.	
Columbia University <i>Graduate Researcher</i>	New York, NY Aug 2025 – Dec 2025
<i>Bayesian Modeling of Collatz Stopping Times – arXiv:2603.04479. N. Bonacorsi & M. Bordoni</i> - Modeled Collatz stopping times over integers up to 10 million using a scalable hierarchical Bayesian negative-binomial specification with partial pooling across modulo-8 residue classes, capturing overdispersion and right-tail heterogeneity. - Calibrated a Monte Carlo simulation engine via Bayesian updating; regime conditioning reduced Wasserstein distance to the empirical distribution by 3.2×, improving right-tail fidelity, posterior calibration, and overall distributional fit.	

Quantitative Research Projects

Bootstrap-Calibrated Distributionally Robust Allocation [GitHub] <i>Independent Quantitative Research</i>	New York, NY May 2026 – Jul 2026
- Constructed a distributionally robust allocation pipeline in Python for 25 Fama–French size–value portfolios, calibrating a covariance ambiguity set via circular-block bootstrap resampling and solving capped long-only quadratic programs with a custom active-set solver. - Achieved an 18.4% reduction in out-of-sample annualized volatility versus equal weight over a 2013–2026 walk-forward evaluation (16.45% vs. 20.15%), with all model selection restricted to 2000–2012 data; delivered a 12.90% CAGR and 0.82 net Sharpe after 10 bps transaction costs.	
Maturity-Safe Distributional Volatility Forecasting [GitHub] <i>Independent Quantitative Research</i>	New York, NY May 2026 – Jul 2026
- Combined six VIX, HAR-style, and EWMA variance models in Python using bounded-loss Hedge updates applied only after 22-day targets matured; calibrated a 90% upper variance forecast from 22 interlaced residual streams with non-overlapping targets within each stream. - Improved out-of-sample volatility forecasting over 2013–2026, lowering log-RMSE by 14.9% versus VIX-implied variance (HAC-robust $p < 0.001$) and achieving 89% coverage at a nominal 90% level across 3,381 forecasts; cost-aware SPY controller delivered 8.3% annualized volatility, 0.84 excess-return Sharpe ratio, and 11.6% maximum drawdown after 1 bp costs.	

Work Experience

E.PTY Group Quantitative Research & Strategy	Bologna, Italy Jun 2023 – Jun 2025
- Built an end-to-end probabilistic decision engine for resource allocation, combining hierarchical demand forecasting, posterior-predictive simulation, and Monte Carlo scenario generation to propagate uncertainty into constrained pricing and budget-allocation decisions; reduced rolling out-of-sample RMSE by 14.6% versus historical forecasting baselines. - Designed and backtested stochastic allocation policies under budget/capacity constraints with explicit downside-risk control and rolling re-optimization as new demand signals arrived; increased realized operating profit by 11.8% while reducing 95% downside CVaR by 17.3% versus the historical rule-based allocation policy.	

Skills

Programming & Data: Python, SQL, R, MATLAB; NumPy, pandas, SciPy, scikit-learn, PyTorch, statsmodels.
Statistical & ML Methods: Time Series, Bayesian Inference, Monte Carlo, GLMs, Statistical Learning, Neural Networks, Model Validation, Risk Modeling.
Mathematics: Stochastic Processes, Stochastic Control, Optimal Stopping, HJB Equations, Optimization, Numerical Methods, PCA/SVD, Information Theory, Uncertainty Quantification.
Languages: Italian (native), English (fluent).